

SIA AI HACKATHON TEAM 8

Use Case – ABC – AI BIM Coordinator

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PROBLEM STATEMENT

1

Corenet X mandates fully coordinated BIM models, increasing architects' responsibility as main coordinators.

2

Traditional BIM coordination relies on exporting models to Navisworks for clash detection.

3

Navisworks lacks a focused methodology, generating excessive false positives that require manual filtering.

4

Reviewing and resolving clashes is repetitive, time-consuming, and diverts focus from major design issues.

5

Current workflows treat all clashes indiscriminately, often including human modeling errors, while failing to highlight critical design issues like headroom constraints.

6

Tens of thousands of unstructured clashes overwhelm BIM coordinators, slowing coordination and decision-making.

SCOPE & OBJECTIVES

A minimum viable product will be prototyped to show AI can be used to identify high impact clashes (coordination issues) that will inhibit design progress. This tool aims to:

1

Process raw data output from Navisworks to high priority actionable items, for coordination.

2

Pinpoint or highlight design issues for resolution.

3

Be used as a BIM Model Quality Checker tool for internal quality checks.

APPROACH & TOOLS

P

Prioritization: Filters out false positives, low-priority clashes, and minor modeling errors, focusing on high-impact design issues that require attention.

A

Assignment: Based on rules defined by designed prompts, Gemini assigns the clashes to the disciplines, then providing an instruction on how to resolve the clash.

I

Identification: Based on the output of the clash report, the clashing item is highlighted and isolated for efficient identification.

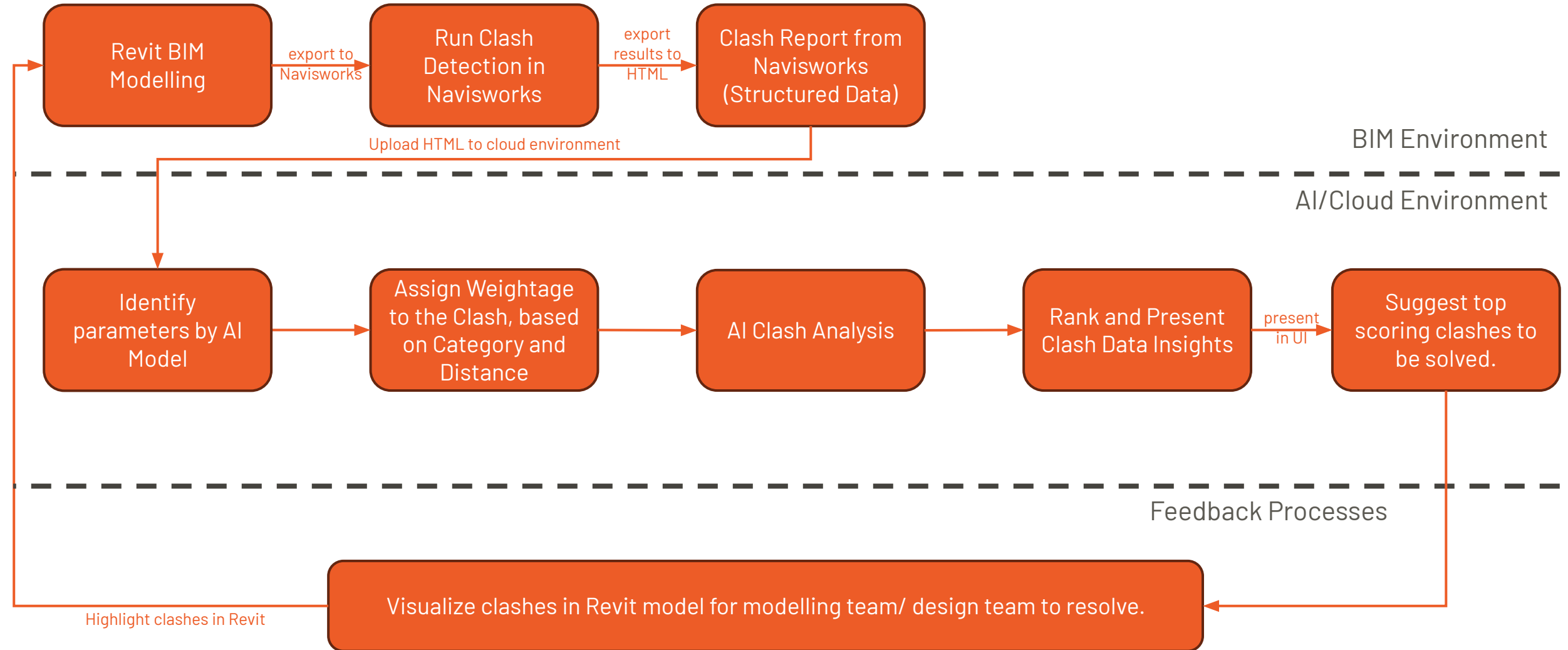
R

Resolution: Project teams take the instruction suggested by Gemini to resolve the clash highlighted.

Data Availability

Data will reside locally, within company domain, as most project data are usually confidential.

STRATEGY: PROPOSED POC WORKFLOW



STRATEGY: CHALLENGES + LIMITATIONS

Data Quality & Consistency

BIM models vary in standards, accuracy, and completeness, making it difficult to train AI effectively.

Distinguishing Subjective Issues

Training AI to correctly accurately distinguish between critical design issues and minor modelling issues could involve processes more advanced than scoring and prioritization.

Not Platform Agnostic

Current MVP is based off Autodesk products, which leads to dependency to a single platform.

Data Collection for AI Improvement Feedback Loop

Project data are confidential, which leads to issues when collecting feedback data for AI Improvement and training.

Scalability & Performance

Processing large, detailed federated models efficiently without excessive computational overhead.

Nature of Clash Detection Process

Clash Detection Process is not able to detect coordination issues like low headroom or modelling issues like floating BIM elements.

TESTS & MEASURES

False Positive Rate Analysis

Compare AI-filtered clashes with traditional Navisworks outputs to measure reduction in irrelevant clashes.

Metric: Reduction in irrelevant clashes (%)

Scalability Test

Run AI on small, medium, and large BIM models to assess performance under different workloads.

Metric: AI processing time across different model sizes (seconds/minutes per model)

Clash Prioritization Validation

Assess whether AI correctly ranks high-impact clashes over minor issues based on expert reviews.

Metric: Accuracy of AI clash ranking (% alignment with expert reviews)

BIM Coordinator Feedback Loop

Conduct user testing with BIM professionals to evaluate ease of use and accuracy of insights.

Metric: User satisfaction score (% positive feedback) and accuracy of AI-generated insights

Processing Speed Benchmark

Measure time taken for AI analysis versus manual clash filtering to evaluate efficiency gains.

Metric: Time reduction in clash filtering (minutes/hours per project)

Task Completion Time Study

Compare time taken for design teams to resolve clashes using AI versus traditional workflows.

Metric: Time reduction for clash resolution (hours saved per project)

OUTCOMES: EXPECTED BENEFITS

Reduced Manual Effort

Automates filtering of clash data, freeing up valuable time from repetitive tasks.

Prioritized Clashes

Uses AI to rank high-impact clashes, ensuring design teams focus on critical issues.

Enhanced Quality Control

Improves BIM model coordination by early detection of errors and headroom issues.

Cost Savings

Minimizes rework during construction, reducing delays and associated expenses.

Regulatory Compliance

Streamlines submissions for portals like Corenet X by ensuring fully coordinated BIM models.

Improved Collaboration

Facilitates seamless multidisciplinary review and coordination through actionable insights.

OUTCOMES: USER BASE

AEC Firms

- Reduces clash detection time, allowing architects and engineers to focus on design improvements.
- Improves interdisciplinary coordination by prioritizing critical clashes over false positives.
- Enhances BIM model quality, reducing costly errors during construction.

Clients / Developers

- Ensures fully coordinated designs before construction, minimizing project delays and budget overruns.
- Improves design quality, leading to higher-value assets with fewer post-construction modifications.
- Increases transparency in the coordination process, improving confidence in project delivery.

Contractors

- Reduces on-site rework and material wastage by resolving major clashes during coordination.
- Improves constructability insights by identifying potential site conflicts early.
- Streamlines communication between project stakeholders through better clash classification and reporting.

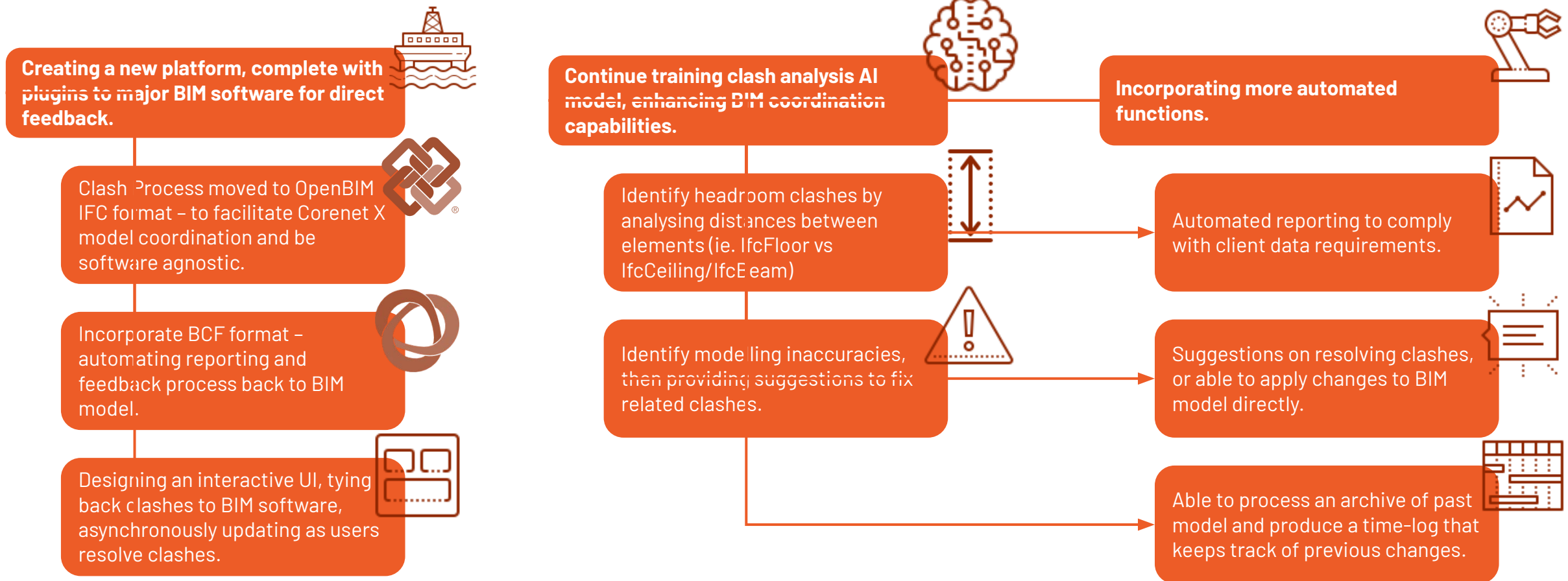
BIM Consultancies

- Enhances BIM coordination services by leveraging AI for faster and more accurate clash analysis.
- Enables consultants to provide higher-value insights rather than manually filtering thousands of clashes.
- Supports automation in BIM audits, improving compliance with regulatory and project standards.

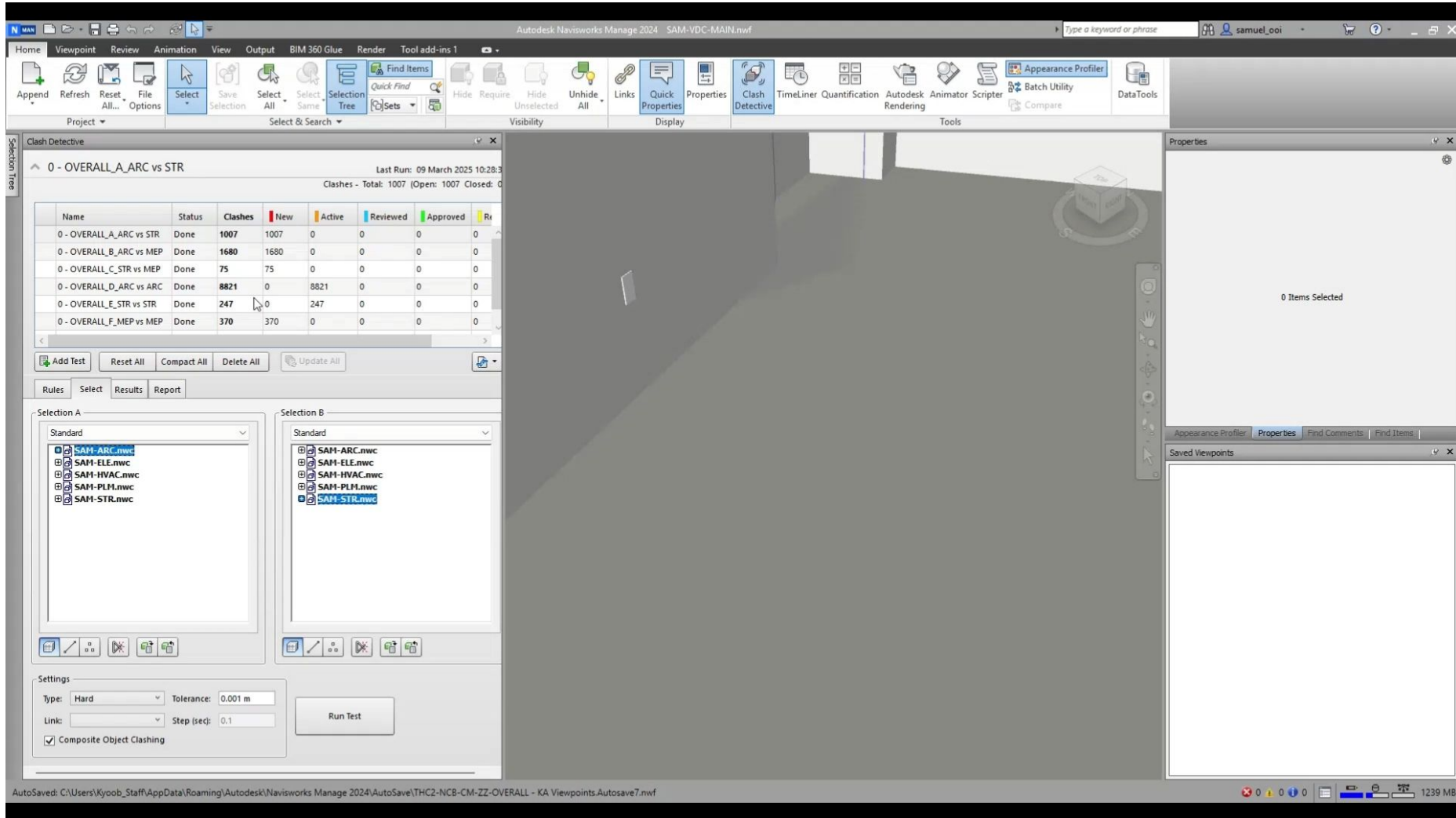
Singapore Government

- Facilitates Corenet X adoption by ensuring BIM models meet full coordination requirements.
- Improves industry-wide BIM efficiency, supporting Singapore's digital construction initiatives.
- Enhances regulatory review processes by reducing uncoordinated submissions and revision cycles.

WHAT'S NEXT: FUTURE DEVELOPMENT



DOCUMENTATION



Video of the POC in Action

SIA AI Hackathon

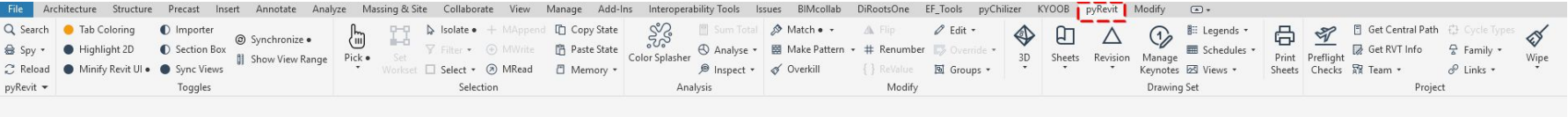
Team 8 Use Case – AI Assisted BIM Coordination

DOCUMENTATION: WORKFLOW WALKTHROUGH

A. Prerequisites:



Project File in Revit



PyRevit with Routes Enabled

AUTODESK®
NAVISWORKS™

Clash Report

Test 1

Image	Clash Name	Status	Distance	Grid Location	Description	Date Found	Clash Point	Item 1 Element Id	Item 1 Element Category	Item 2 Element Id	Item 2 Element Category
	Clash1	New	-0.180		Hard	2025/5/18 06:25	x:417638.821, y:78725.030, z:251.208	1538999	Electrical Equipment	793240	Structural Framing
	Clash2	New	-0.141		Hard	2025/5/18 06:25	x:417638.889, y:78725.063, z:247.144	1538413	Electrical Equipment	792353	Structural Framing
	Clash3	New	-0.057		Hard	2025/5/18 06:25	x:417626.495, y:78730.184, z:236.842	1482764	Electrical Fixtures	670327	Floors
	Clash4	New	-0.057		Hard	2025/5/18 06:25	x:417624.714, y:78729.315, z:236.842	1482687	Electrical Fixtures	670327	Floors
	Clash5	New	-0.019		Hard	2025/5/18 06:25	x:417639.904, y:78723.542, z:251.105	1538999	Electrical Equipment	793010	Structural Framing
	Clash6	New	-0.019		Hard	2025/5/18 06:30	x:417639.226, y:78733.103, z:247.460	1538413	Electrical Equipment	792120	Structural Framing

Navisworks Clash Detection Report HTML

Clash AI Settings

File Edit View Insert Format Data Tools Extensions Help

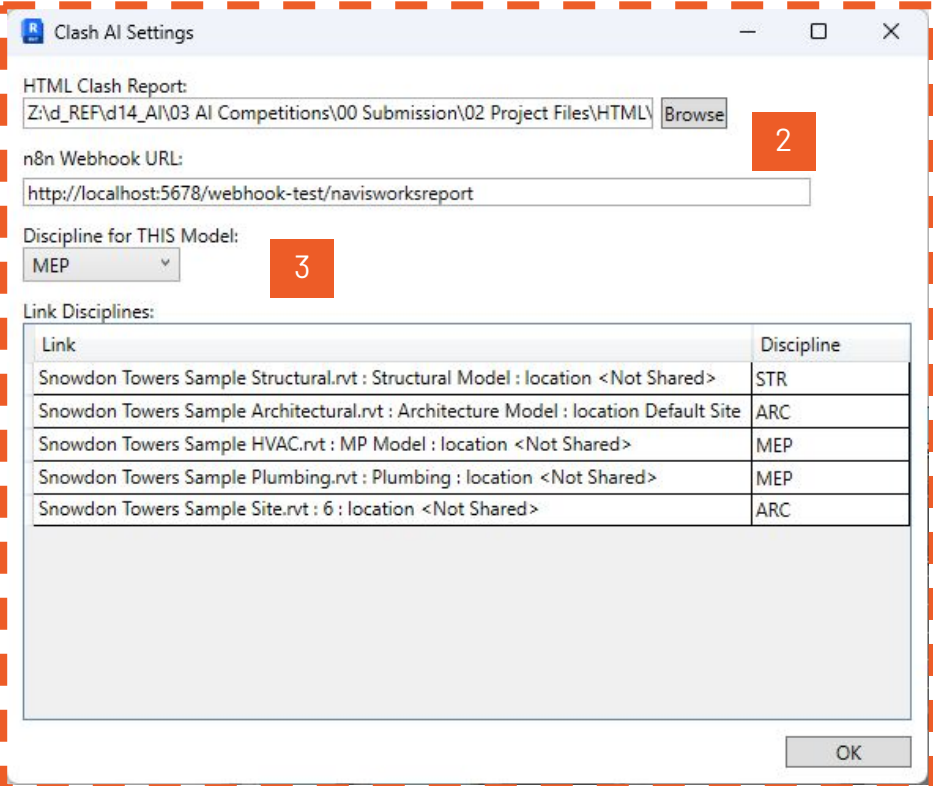
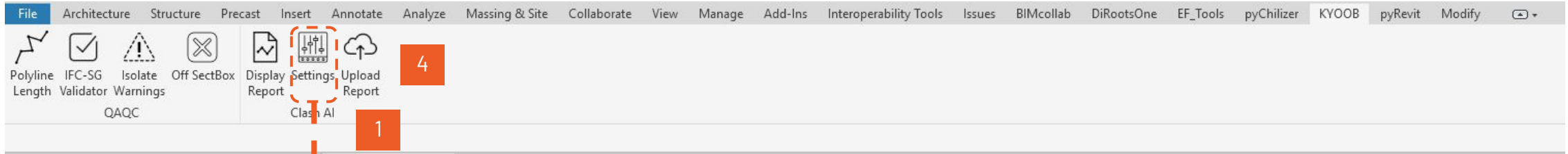
Menus 100% 123

A1	Category	Score	Discipline	B	C	D	E	F	G
1	Category								
2	Air Terminals	8	MEP						
3	Analytical Beams	0							
4	Analytical Columns	0							
5	Analytical Floors	0							
6	Analytical Foundation Slabs	0							
7	Analytical Isolated Foundations	0							
8	Annotations	0							
9	Areas	0							
10	Balusters	2	ARC						
11	Boundary Conditions	0							
12	Cable Tray Fittings	8	MEP						
13	Cable Trays	8	MEP						
14	Casework	3	ARC						
15	Ceilings	6	ARC						
16	Color Fill Legends	0							
17	Columns	9	ARC						
18	Communication Devices	2	MEP						
19	Conduit Fittings	5	MEP						
20	Conduits	5	MEP						
21	Curtain Panels	5	ARC						
22	Curtain Systems	5	ARC						
23	Curtain Wall Mullions	5	ARC						
24	Data Devices	2	MEP						
25	Detail Items	0							
26	Dimensions	0							
27	Division Profiles	0							
28	Doors	6	ARC						
29	Duct Accessories	10	MEP						
30	Duct Color Fill Legends	0							
31	Duct Fittings	10	MEP						

User Configuration of Categories, Clash Score and Discipline on Google Sheets

DOCUMENTATION: WORKFLOW WALKTHROUGH

A. Revit Settings Setup:



With PyRevit, we have scripted some tools with the help of Gemini to interface the clash data with Gemini.

1: Pull up the settings configuration.

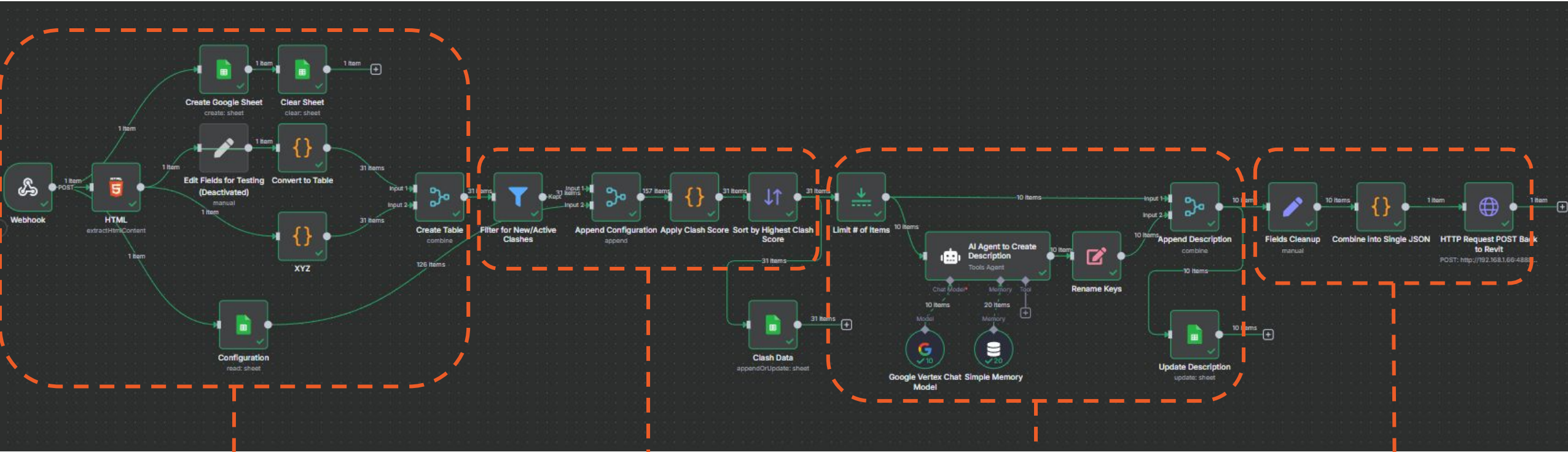
2: Browse for the HTML to be uploaded to n8n for processing, enter the webhook URL in n8n.

3: Set up the disciplines for the current model and link models. This is to identify which model that we are working with and which link models to isolate later.

4: Once completed, click on OK and then 'Upload Report'. The report will be uploaded to n8n via HTTP POST request.

DOCUMENTATION: WORKFLOW WALKTHROUGH

B. Clash Data Processing on n8n:



Parsing Data

Clash data from Navisworks clash report is converted to json arrays.

P: Prioritization

Clash score is assigned based on the category / severity of the clash.

A: Assignment

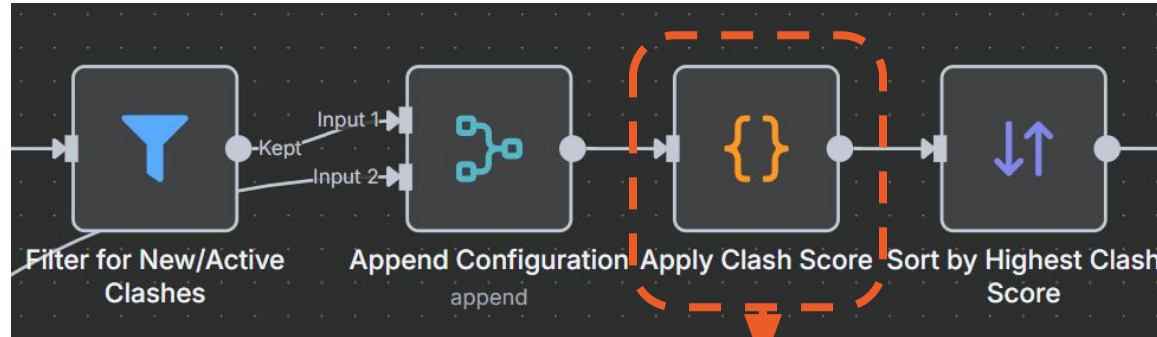
Gemini analyses the clash data and produces a targeted output.

POSTing Data back to Revit

The data produced here is posted back to Revit for further action

DOCUMENTATION: WORKFLOW WALKTHROUGH

PRIORITIZATION DEEP DIVE



$$\text{Clash Score} = (\text{Element 1 Score} * \text{Element 2 Score}) * \text{Severity Factor}$$

Where:

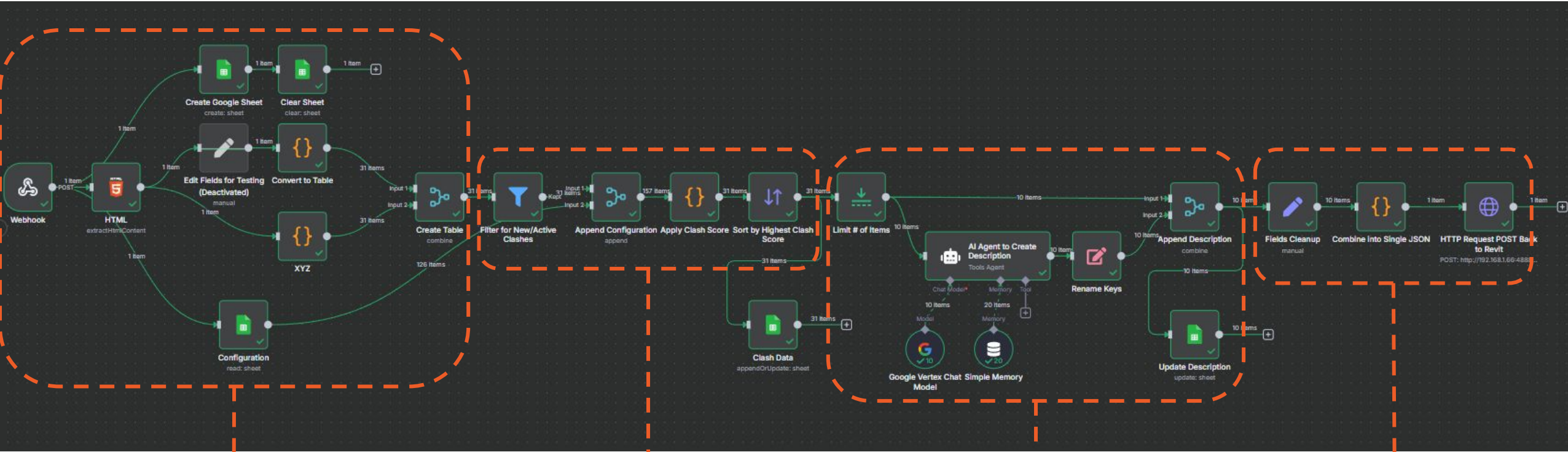
- Element Score: Based on category weightage.
(e.g., Structural = 10, MEP = 8, Architecture = 4, Fixtures = 1)
- Severity Factor(User Configuration):

Severity	Multiplier	Condition
Severe	4	Penetration $\geq$ 100 mm
High	2	50 mm $\leq$ Penetration < 100 mm
Medium	1.2	10 mm $\leq$ Penetration < 50 mm
Low	0.5	Penetration < 10 mm

Penetration = |Distance| $\times$ 1000 (in mm), Default construction tolerance is 50mm.

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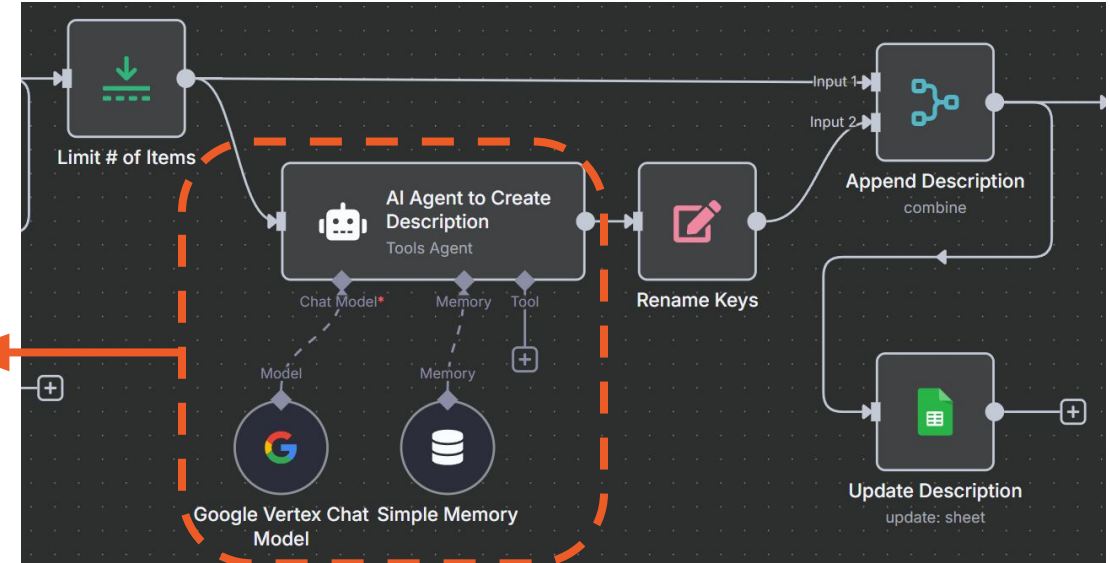
ASSIGNMENT DEEP DIVE

Task Prompt:

Use the below format to describe the clash: A/An {{ \$json.element_1_cat }} element clashes with a {{ \$json.element_2_cat }} element, with a penetration distance of {{ \$json.distance }} mm, (replace {{ \$json.severity_factor }} with the description below.). The clash is assigned to (which discipline to amend the clash). (Suggest a way to amend this clash.){ {{ \$json.discipline_1 }} and {{ \$json.discipline_2 }} are the category disciplines for the clash. MEP is most likely to attend to the clash, then ARC, then lastly STR. Assign the clash based on this priority.{{ \$json.severity_factor }} means how serious is the clash, 4 is 'severe', 2 is 'high', 1.2 is 'medium' and 0.5 is 'low'. Make the description sound organic, avoid the use of ':'.

System Prompt:

You are an AI assistant designed to produce descriptions for each of the clash in Navisworks. {{ \$json.element_1_cat }} means the category of the first item; {{ \$json.element_2_cat }} means the category of the second item; {{ \$json.distance }} means how much the items are intersecting each other, in meters, to convert to mm. {{ \$json.severity_factor }} means how serious is the clash, 4 is severe, 2 is high, 1.2 is medium and 0.5 is low. {{ \$json.discipline_1 }} and {{ \$json.discipline_2 }} are the category disciplines for the clash. MEP is most likely to attend to the clash, then ARC, then lastly STR. Assign the clash based on this priority.

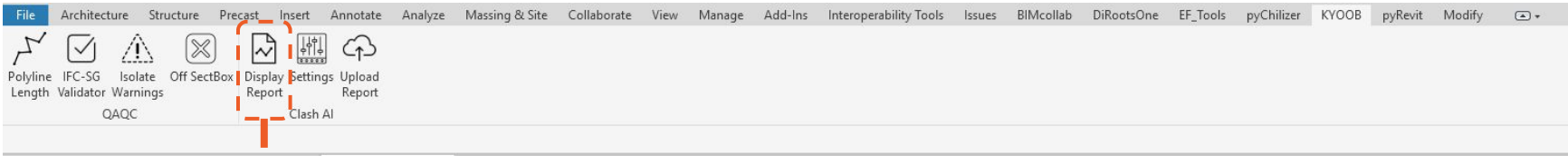


Sample description generated:

An Electrical Equipment element clashes with a Structural Framing element, with a penetration distance of -180 mm, and is considered a severe clash. The clash is assigned to MEP. Relocate the electrical equipment to avoid the structural framing.

DOCUMENTATION: WORKFLOW WALKTHROUGH

C. Displaying Report on Revit:



I: Identification

Each clash comes with an isolate button. This will isolate the clashing element in the model, providing instant identification of the clash area.

Isolate	Clash Name	Clash Score	Description	Status	Distance	Date Found	Element 1 Id	Element 1 Cat	Discipline 1	Element 2 Id	Element 2 Cat	Discipline 2	X	Y	Z
Isolate	Clash1	180	A Electrical Equipment element clashes with a Structural Framing element, with a penetration distance of -180 mm, and is considered a severe clash. The clash is assigned to MEP. Relocate the electrical equipment to avoid the structural framing.	New	-0.180	2025/5/18 06:25	1538999	Electrical Equipmen	MEP	793240	Structural Framing	STR	417638.821	78725.030	251.208
Isolate	Clash2	180	An Electrical Equipment element clashes with a Structural Framing element, with a penetration distance of -141 mm, and is considered a severe clash. The clash is assigned to MEP. Relocate the electrical equipment to avoid the structural framing member, or adjust the framing layout if possible.	New	-0.141	2025/5/18 06:25	1538413	Electrical Equipmen	MEP	792353	Structural Framing	STR	417638.889	78725.063	247.144
Isolate	Clash5	54	An Electrical Equipment element clashes with a Structural Framing element, with a penetration distance of -19 mm, and is considered a medium clash. The clash is assigned to MEP. Slightly adjust the position of the electrical equipment to clear the structural framing.	New	-0.019	2025/5/18 06:25	1538999	Electrical Equipmen	MEP	793010	Structural Framing	STR	417639.904	78723.542	251.105
Isolate	Clash6	54	An Electrical Equipment element clashes with a Structural Framing element, with a penetration distance of -19 mm, and is considered a medium clash. The clash is assigned to MEP. Slightly adjust the position of the electrical equipment to clear the structural framing.	New	-0.019	2025/5/18 06:25	1538413	Electrical Equipmen	MEP	792120	Structural Framing	STR	417639.226	78723.193	247.469
Isolate	Clash11	32.400	An Electrical Fixtures element clashes with a Structural Framing element, with a penetration distance of -11 mm, and is considered a medium clash. The clash is assigned to MEP. Slightly adjust the position of the electrical fixture to avoid the structural framing.	New	-0.011	2025/5/18 06:25	1711431	Electrical Fixtures	MEP	630776	Structural Framing	STR	417613.730	78707.315	243.100
Isolate	Clash3	30	An Electrical Fixtures element clashes with a Floor element, with a penetration distance of -57 mm, and is considered a high	New	-0.057	2025/5/18 06:25	1482764	Electrical Fixtures	MEP	670327	Floors	ARC	417626.495	78730.184	236.842

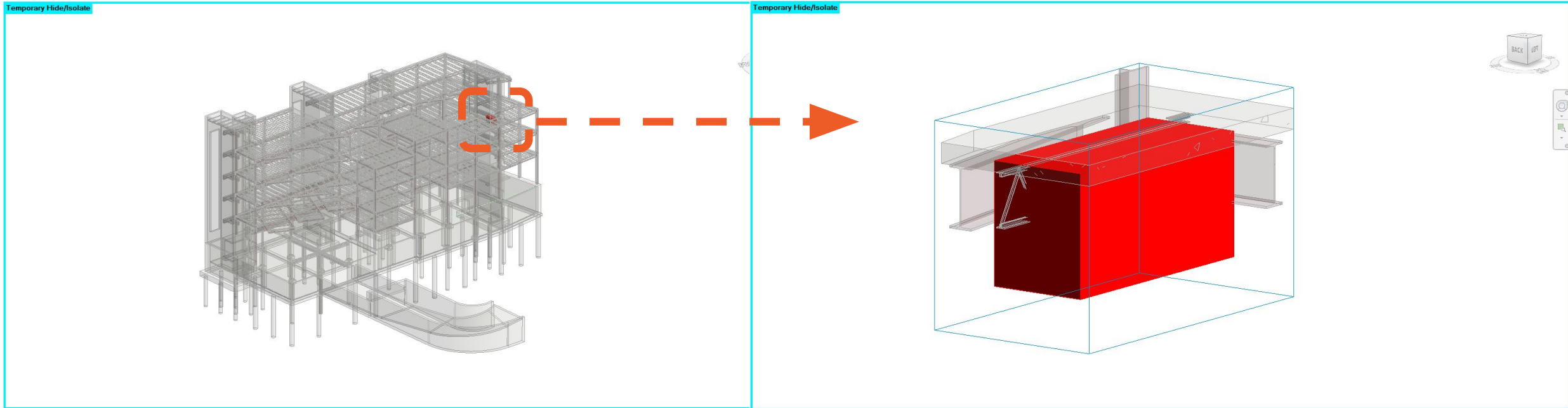
Clash Description from
Google Gemini

Clash Score from n8n

Clash Data from Navisworks

DOCUMENTATION: WORKFLOW WALKTHROUGH

D. Isolating Clash on Revit:



R: Resolution

After identifying the issue, the project team can then work on resolving it, taking suggestions from Gemini's generated description.